

OXIDATIVE PHOSPHORYLATION

OXIDATIVE PHOSPHORYLATION: The formation of ATP from ADP and Pi is termed **Phosphorylation**, as phosphorylation is coupled with **biological oxidation**, the process is called **biological oxidative phosphorylation**.

MECHANISM OF OXIDATIVE PHOSPHORYLATION:

CHEMIOSMOTIC THEORY:

The chemiosmotic coupling hypothesis is the most accepted theory. The chemiosmotic theory which was proposed by the *British biochemist Peter Mitchell* explains the mechanism of **oxidative phosphorylation**. The theory states that the energy released from oxidation generates the **electrochemical potential** by the pumping of protons across the inner mitochondrial membrane and the energy in this electrochemical potential can be converted into ATP. There are three basic principles of the theory.

- 1) The major electron carriers are organized into three complexes, complex I, III and IV (Figure 10.6), which span the inner mitochondrial membrane. The energy released during transport of electrons from one carrier to another allows proton to be pumped across the inner mitochondrial membrane from the matrix to the outside (intermembrane space).
- 2) The inner mitochondrial membrane is impermeable to protons, so that their pumping results in the generation of the **electrochemical potential**.
- 3) Due to this electrochemical potential or proton-motive force, the H^+ ions ejected out by electron transport flow back into the mitochondrial matrix down its electrochemical gradient through **F₀F₁ATPase** molecule (Figure 10.6). The free energy is released as H^+ ion flows back through the **F₀F₁ATPase** into the zone of lower H^+ concentration. The free energy released is coupled with the phosphorylation of ADP to ATP. The **F₀F₁ATPase** catalyzes the addition of inorganic phosphate to ADP to form ATP.

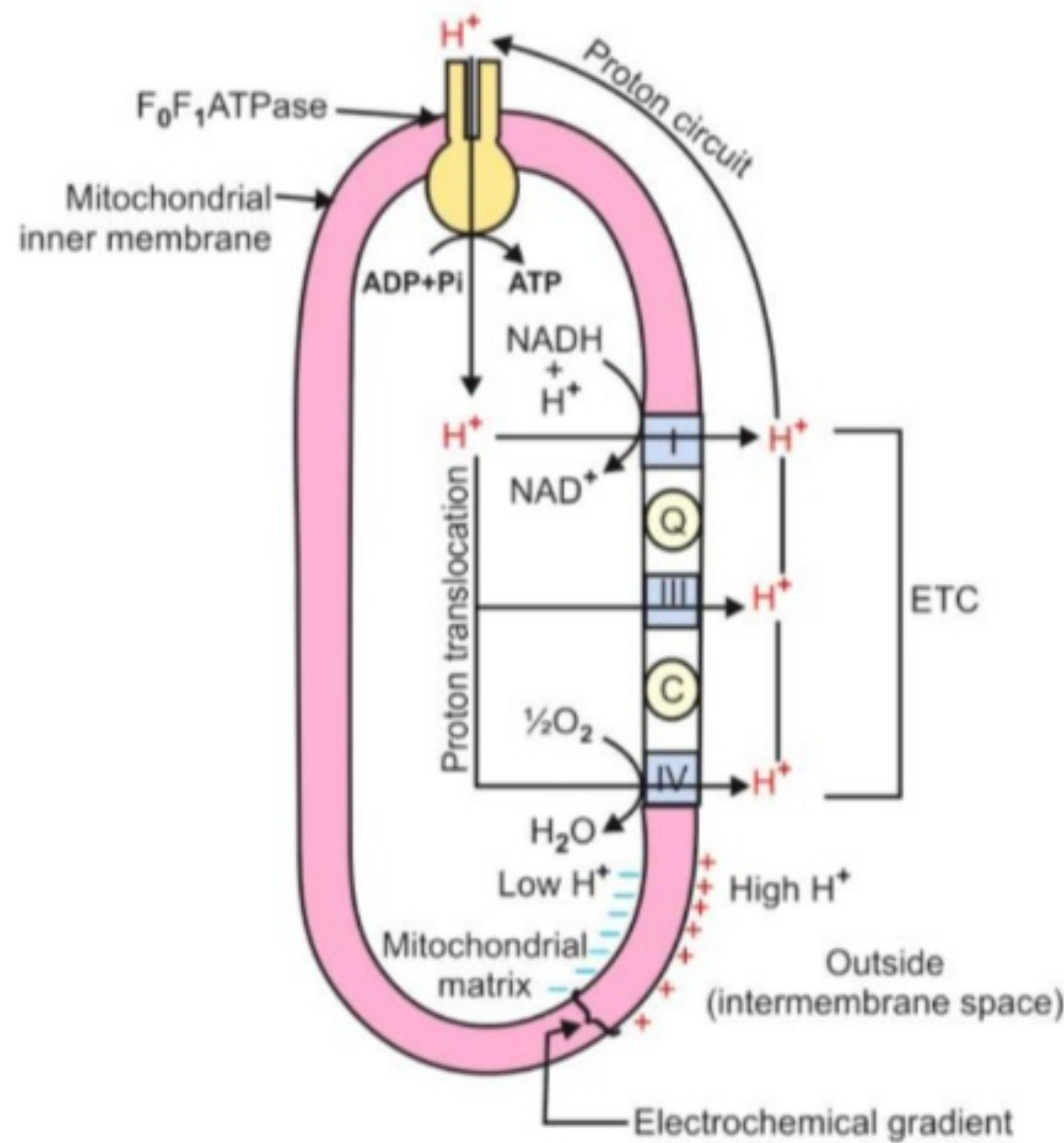


Figure 10.6: Schematic diagram of chemiosmotic theory in which respiratory complexes, I, III and IV act as a proton pump and generate electrochemical gradient

P:O RATION:

- The P:O ratio is a measure of the number of high energy phosphates (i.e. number of ATP molecules) synthesized per atom of oxygen consumed or per molecule of water produced.
- The P:O ratio for oxidation of metabolites that yield NADH is 3 and the ratio for those that yield FADH₂ is 2.

SUBSTRATE LEVEL PHOSPHORYLATION:

The formation of ATP, directly coupled to metabolic process without involvement of electron transport chain and molecular O₂, is called the production of ATP at the substrate level.

Examples of substrate level phosphorylation processes are the conversion of:

- Phosphoenolpyruvate to pyruvate
- 1,3 biphosphoglycerate to 3-phosphoglycerate
- Succinyl-CoA to succinate (Figure 10.7).

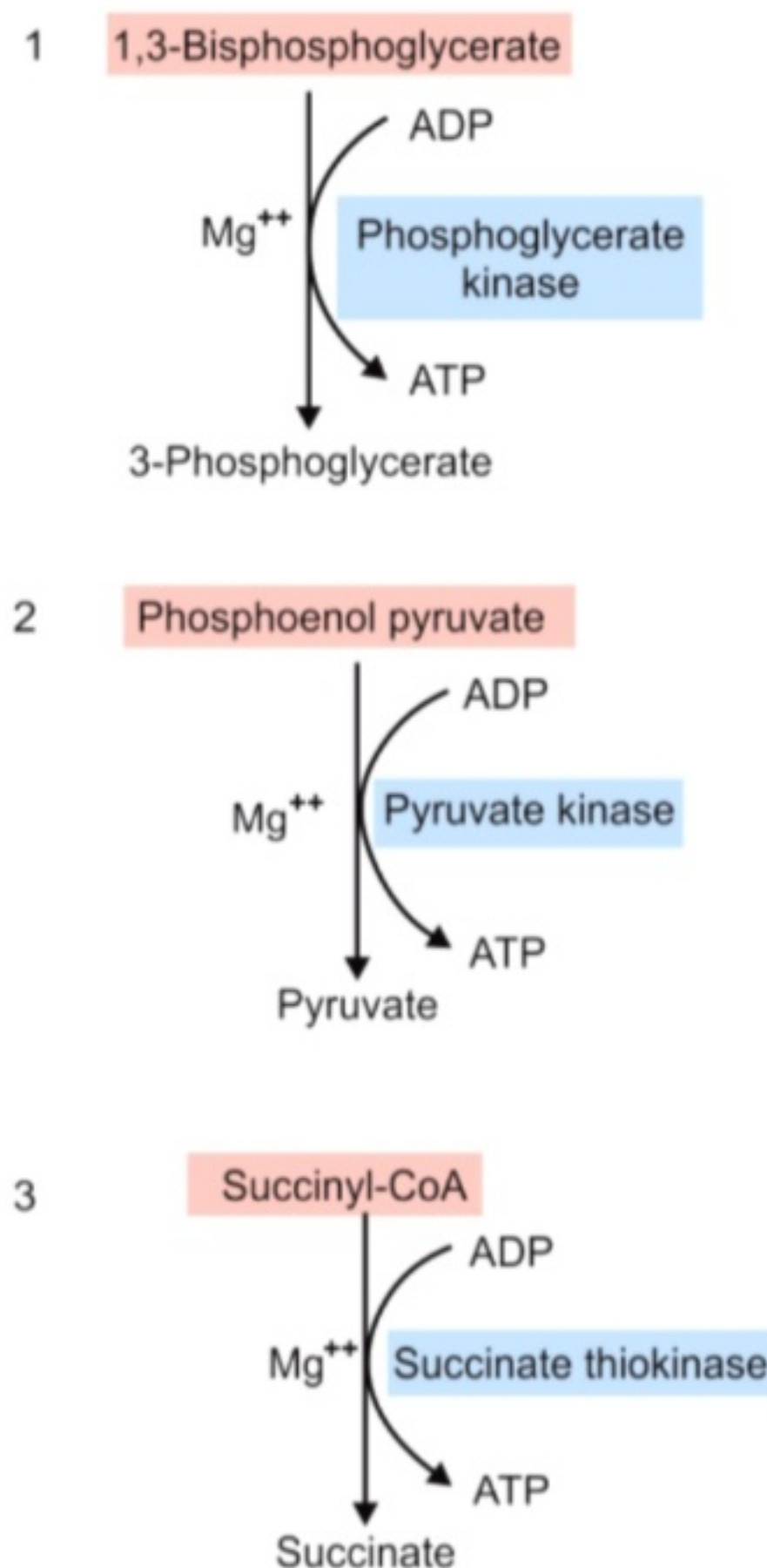


Figure 10.7: Substrate level phosphorylation processes. Reaction 1 and 2 are of glycolysis and 3 occurs in TCA cycle

SHUTTLE SYSTEMS FOR OXIDATION OF EXTRAMITOCHONDRIAL NADH:

- Most of the NADH and FADH₂ entering the mitochondrial electron transport chain arises from Kreb's cycle and β -oxidation of fatty acids, located in the mitochondria itself.
- Since, the inner mitochondrial membrane is not permeable to cytoplasmic NADH, how can the NADH generated by glycolysis, which take place outside of the mitochondria, be oxidized to NAD by respiratory chain located in mitochondria.
- Special shuttle systems carry reducing equivalents from cytosolic NADH (rather than NADH itself) into the mitochondria by an indirect route.
- Two such shuttle systems that can lead to the transport of reducing equivalent from the cytoplasm into mitochondria are:
 - 1) The malate-aspartate shuttle
 - 2) The glycerol phosphate shuttle.

1) THE MALATE-ASPARTATE SHUTTLE SYSTEM:

- The reducing equivalents of cytosolic NADH are first transferred to cytosolic oxaloacetate to yield malate by cytosolic malate dehydrogenase.
- Malate, which carries the reducing equivalents, is transported across the inner membrane by a **dicarboxylate transport** system.
- The reducing equivalents carried by malate are then transferred to mitochondrial NAD⁺ by **mitochondrial malate dehydrogenase** and malate itself gets reoxidized to oxaloacetate.
- The resulting mitochondrial NADH is oxidized by the mitochondrial electron transport chain, leading to formation of 3 molecules of ATP.
- The oxaloacetate so formed cannot pass through the membrane from the mitochondrion back into cytosol, so it is converted to **aspartate by transamination reaction** which is transported to the cytosolic side via amino acid transport system.
- In the cytosol, a reversal of the aspartate aminotransferase reaction gives rise to oxaloacetate and glutamate, thereby completing the “**shuttle like**” process.

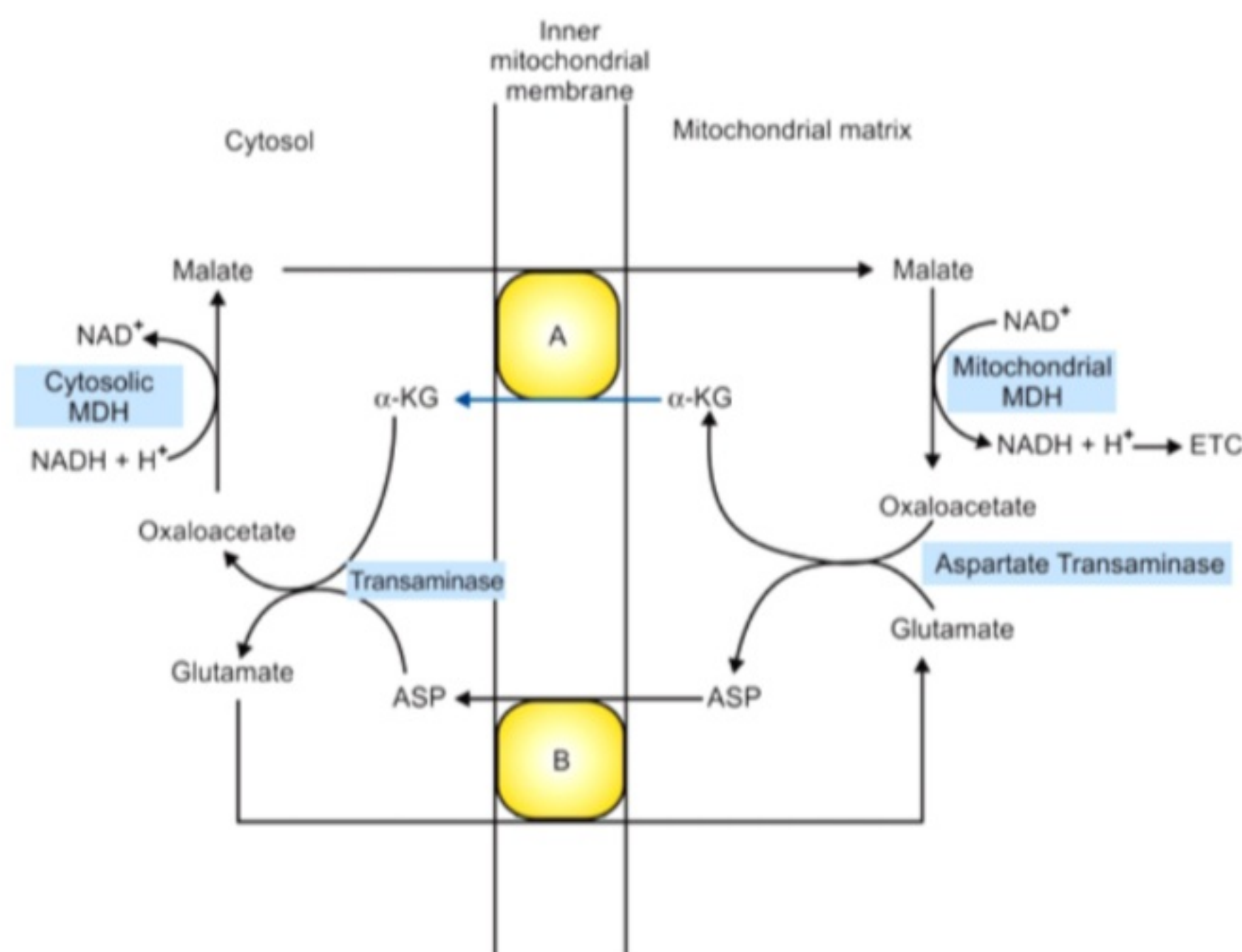


Figure 10.8: Malate-aspartate shuttle

Where	A	:	Malate/α-KG transporter
	B	:	Aspartate/glutamate transporter
	α-KG	:	α-ketoglutarate
	ASP	:	Aspartate
	MDH	:	Malate dehydrogenase
	ETC	:	Electron transport chain

2) GLYCEROL PHOSPHATE SHUTTLE:

- The first step in this shuttle is the transfer of a pair of electrons from cytosolic NADH to dihydroxyacetone phosphate to form glycerol-3-phosphate catalyzed by cytosolic glycerol-3-phosphate dehydrogenase enzyme.
- Glycerol-3-phosphate in turn diffuses through the outer mitochondrial membrane into the intermembrane space of the mitochondria.
- Here glycerol-3-phosphate is reoxidized to dihydroxyacetone phosphate on the outer surface of the inner mitochondrial membrane by a membrane bound FAD containing isoenzyme of glycerol-3-phosphate dehydrogenase.
- An electron pair from glycerol-3-phosphate is transferred to an FAD of the enzyme to form FADH₂. FADH₂ gets oxidized via ETC to generate 2 ATP.
- The dihydroxyacetone phosphate returns to the cytosol and can be reused for reduction of glycerol-3-phosphate.

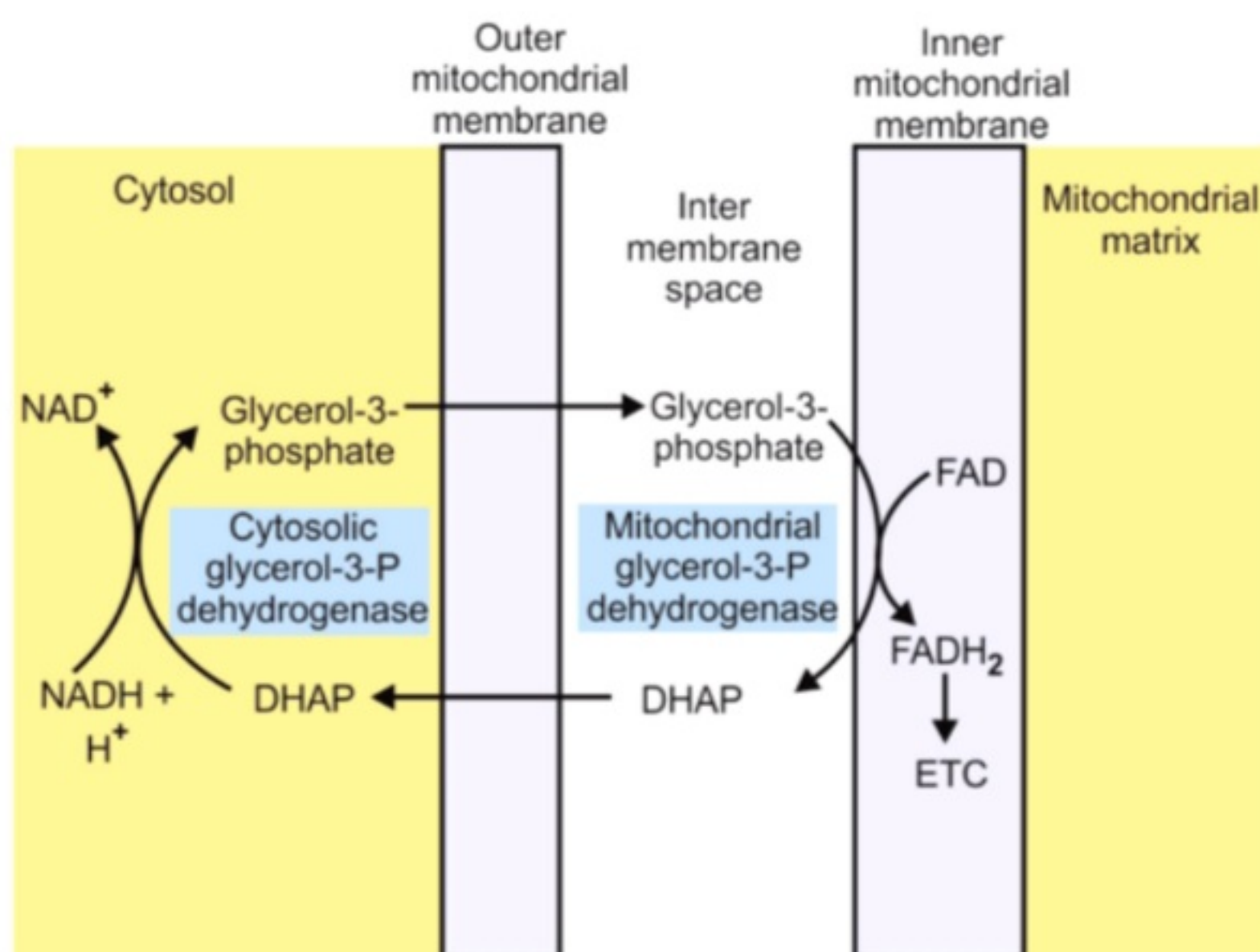


Figure 10.9: Glycerol phosphate shuttle where, DHAP: Dihydroxyacetone phosphate



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